Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Methylene (CH2)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 3
Total e-   : 8
n-body order : 2

GROUND STATE ENERGIES

E0 (Ha) : -38.92428881
E1 (Ha) : -38.87208522
Gap (Ha) : 0.05220359
HF (Ha) : -38.92132899

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,5)

Frozen core orbs : 1
Frozen e-   : 2
Active e-   : 6
Active orbs   : 5
Spin-orbitals   : 10
Hilbert dim   : 100

SPIN PROPERTIES

<S2>   : 2.000000
S   : 1.0000
<Sz>   : 0.000000
Multiplicity : 3
Spin state   : Triplet (S=1)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.997965
<D> / site : 0.399593

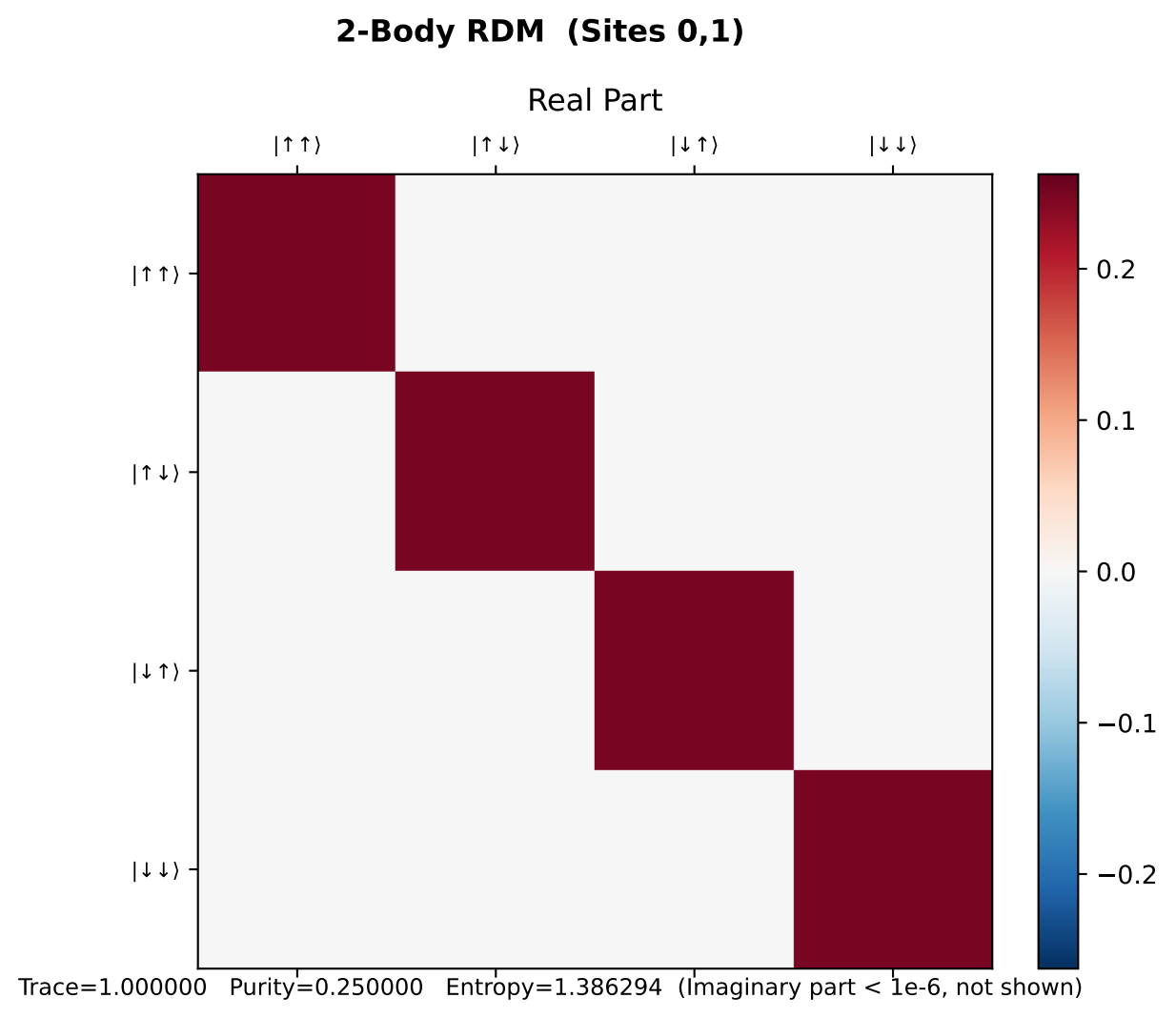
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Body order : 2
Clusters   : 10

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Summary — 2-Body RDMs (Methylene, CAS(6,5))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1	0.250000	—	—	—	—	—	MIXED
0,2	0.250000	—	—	—	—	—	MIXED
0,3	0.250000	—	—	—	—	—	MIXED
0,4	0.299456	—	—	—	—	—	MIXED
1,2	0.250000	—	—	—	—	—	MIXED
1,3	0.250000	—	—	—	—	—	MIXED
1,4	0.250000	—	—	—	—	—	MIXED
2,3	0.995793	—	—	—	—	—	WEAKLY MIXED
2,4	0.306781	—	—	—	—	—	MIXED
3,4	0.360588	—	—	—	—	—	MIXED


2-Body RDM (Sites 0,1)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,1)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

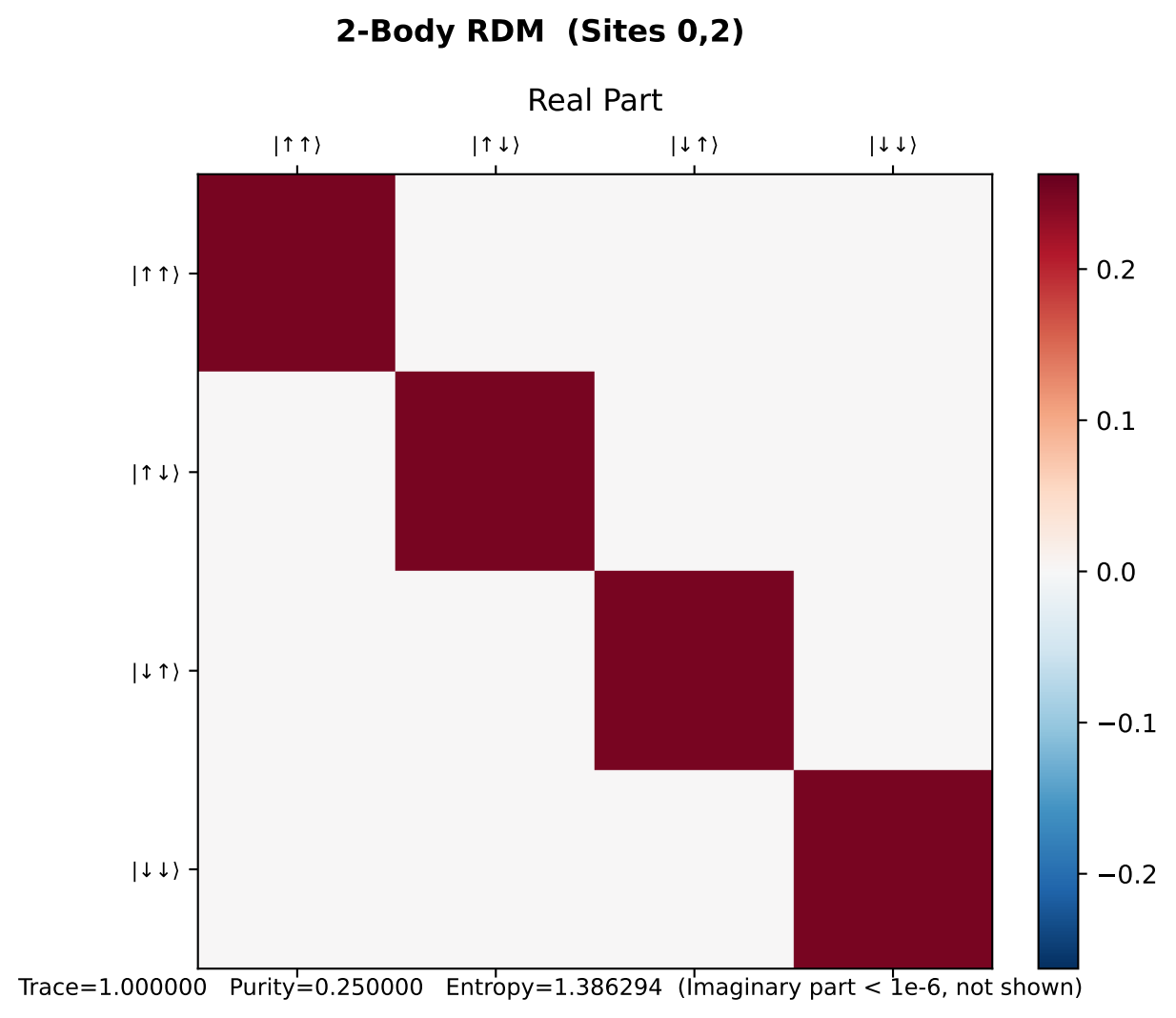
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,2)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500004282
Von Neumann entropy	: 1.3862935050
Eigenvalue min	: 2.497531e-01
Eigenvalue max	: 2.505505e-01
Imaginary part	: < 1e-6 (not shown)

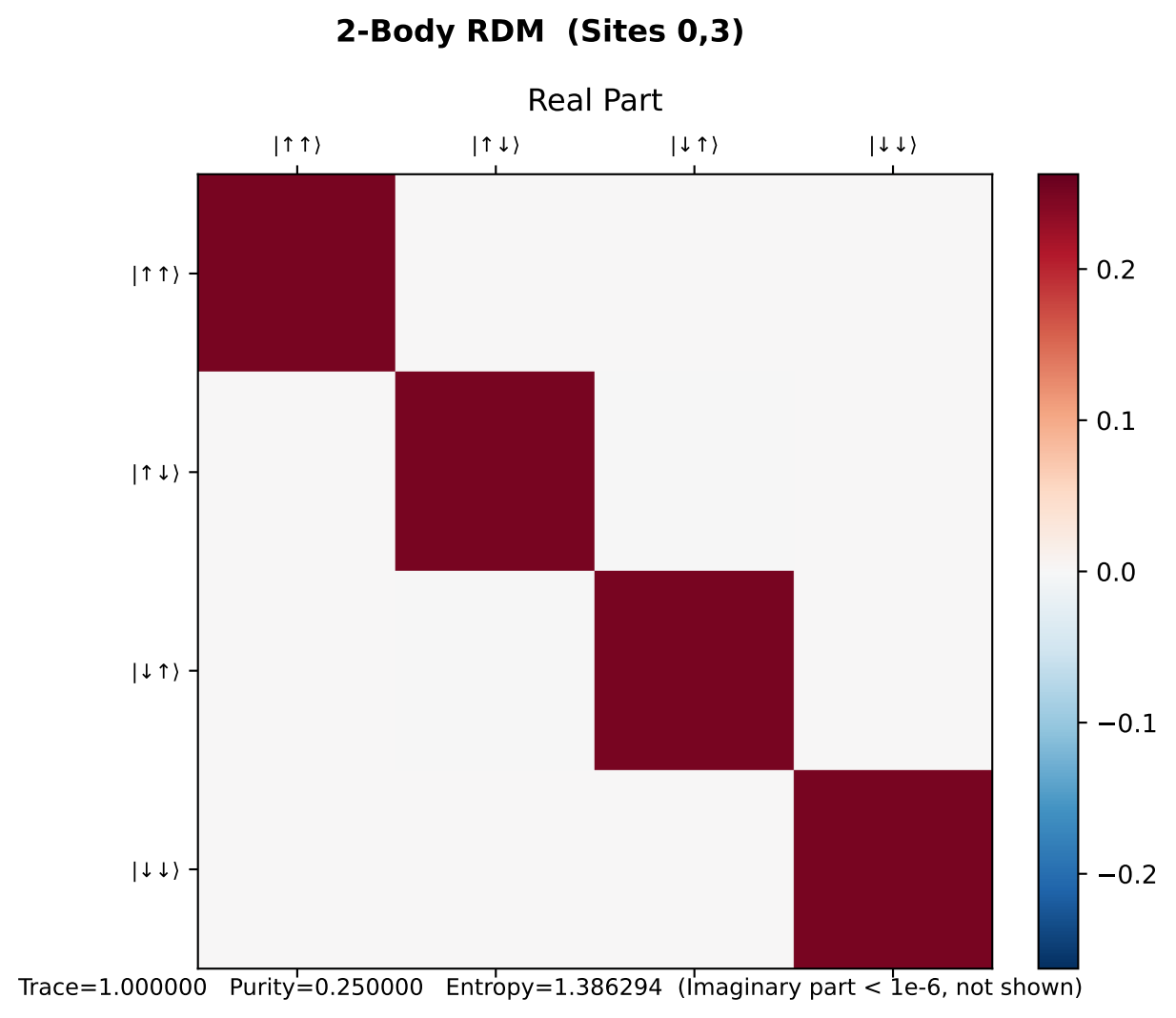
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000043

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500002058
Von Neumann entropy	: 1.3862939495
Eigenvalue min	: 2.497826e-01
Eigenvalue max	: 2.503092e-01
Imaginary part	: < 1e-6 (not shown)

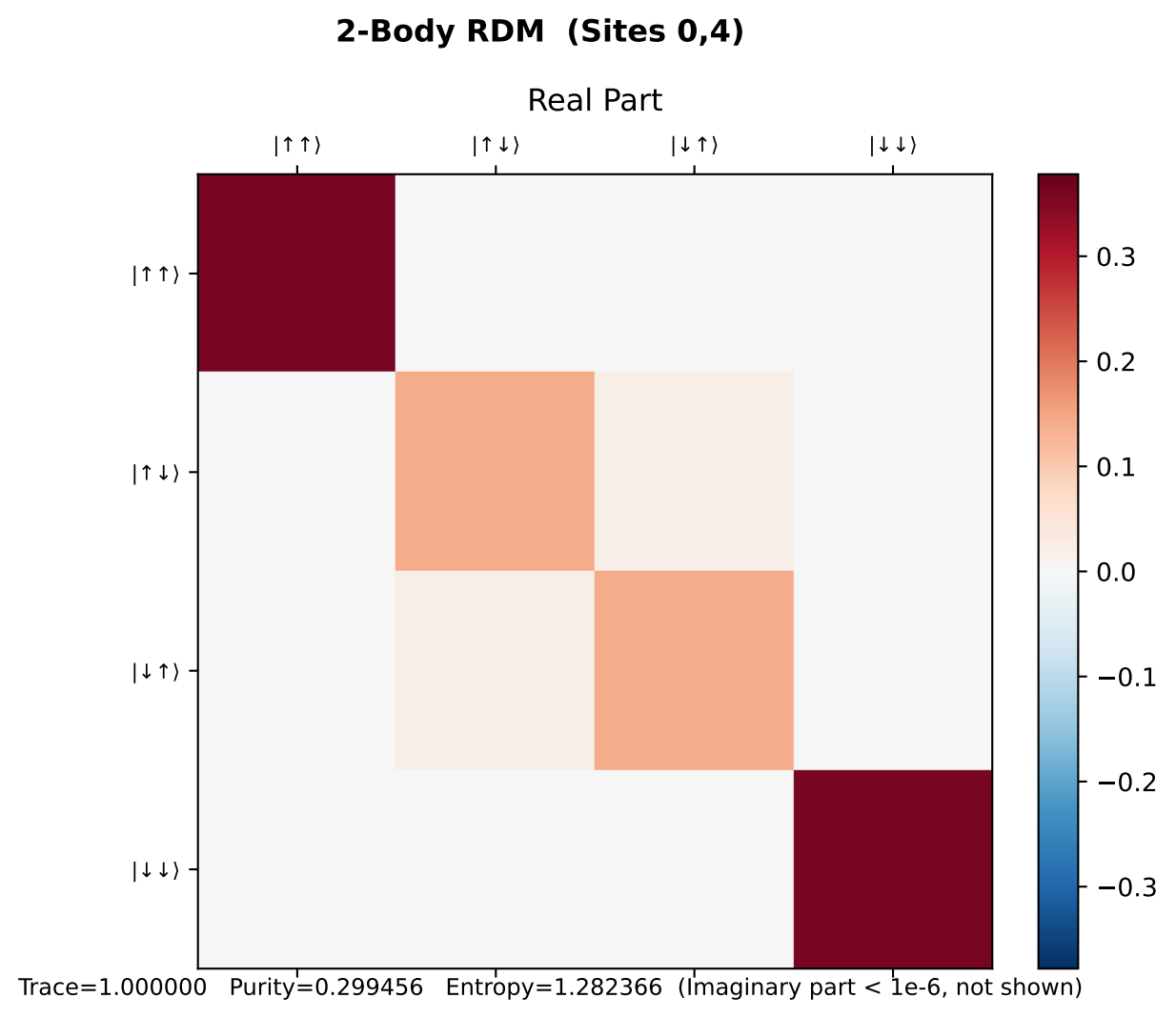
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000021

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 0,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.299456 Entropy=1.282366 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 0,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2994560173
Von Neumann entropy	: 1.2823657791
Eigenvalue min	: 1.173933e-01
Eigenvalue max	: 3.600429e-01
Imaginary part	: < 1e-6 (not shown)

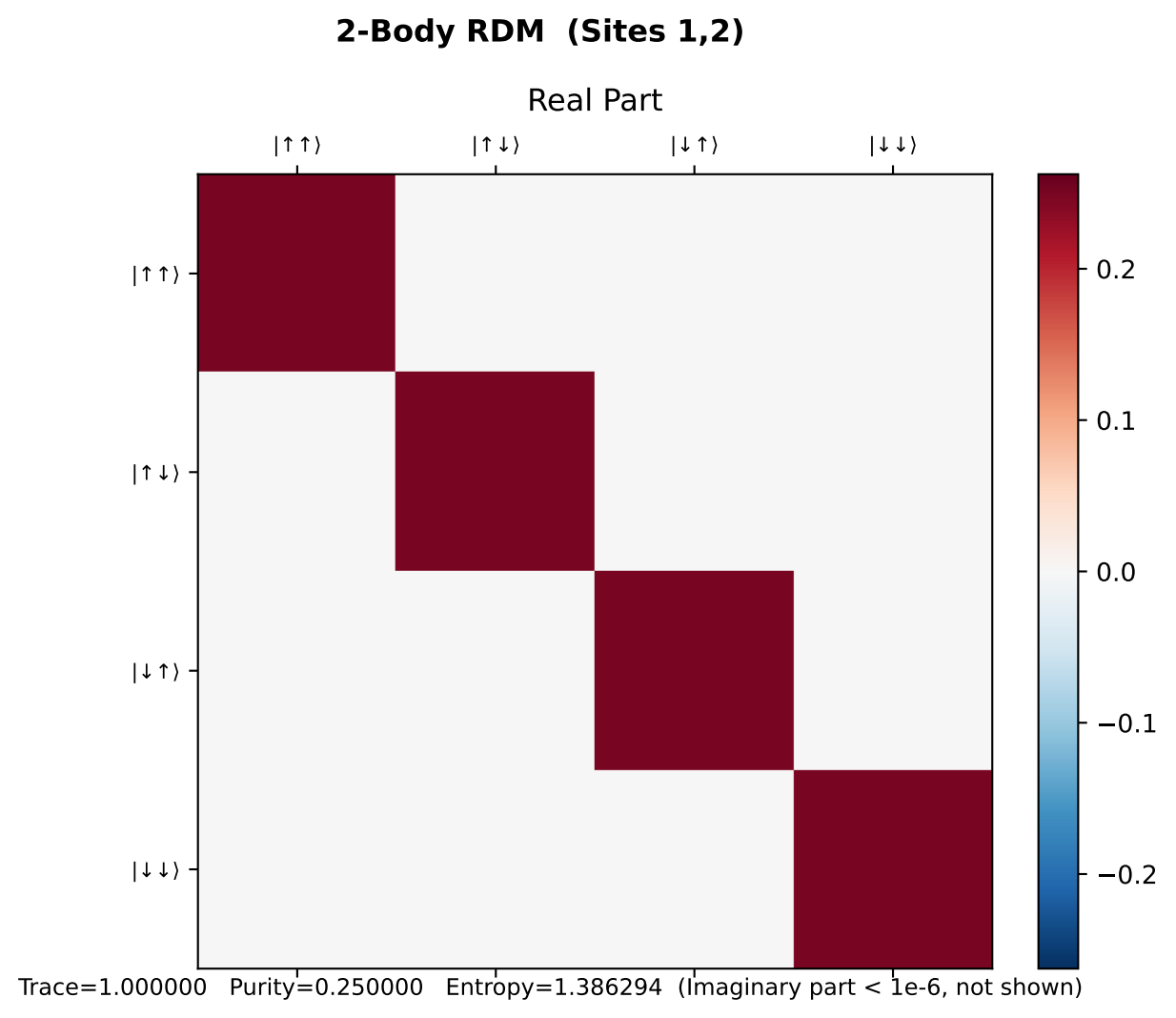
GME Analysis
==========

gamma = Tr(rho^2) = 0.29945602

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,2)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

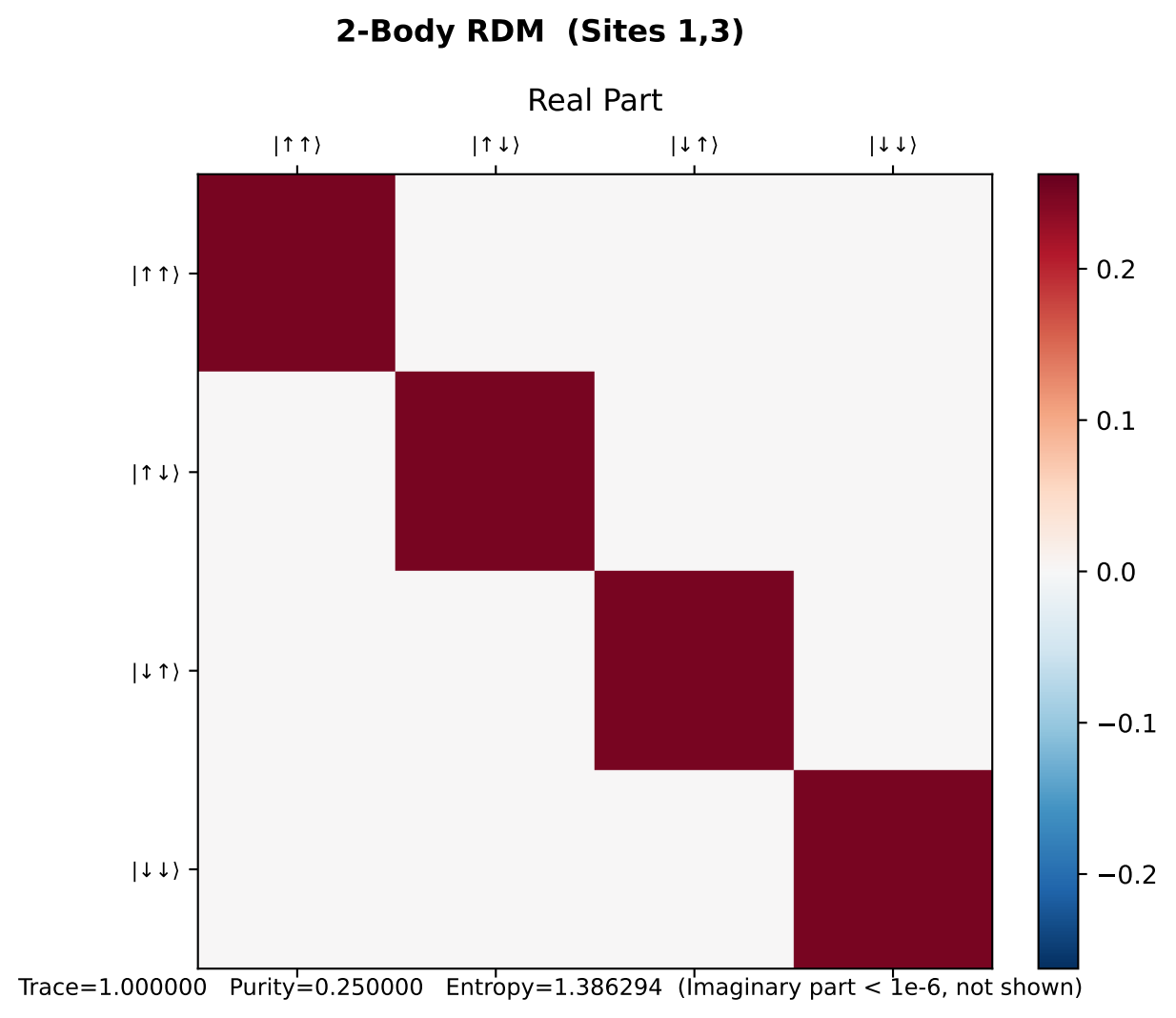
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

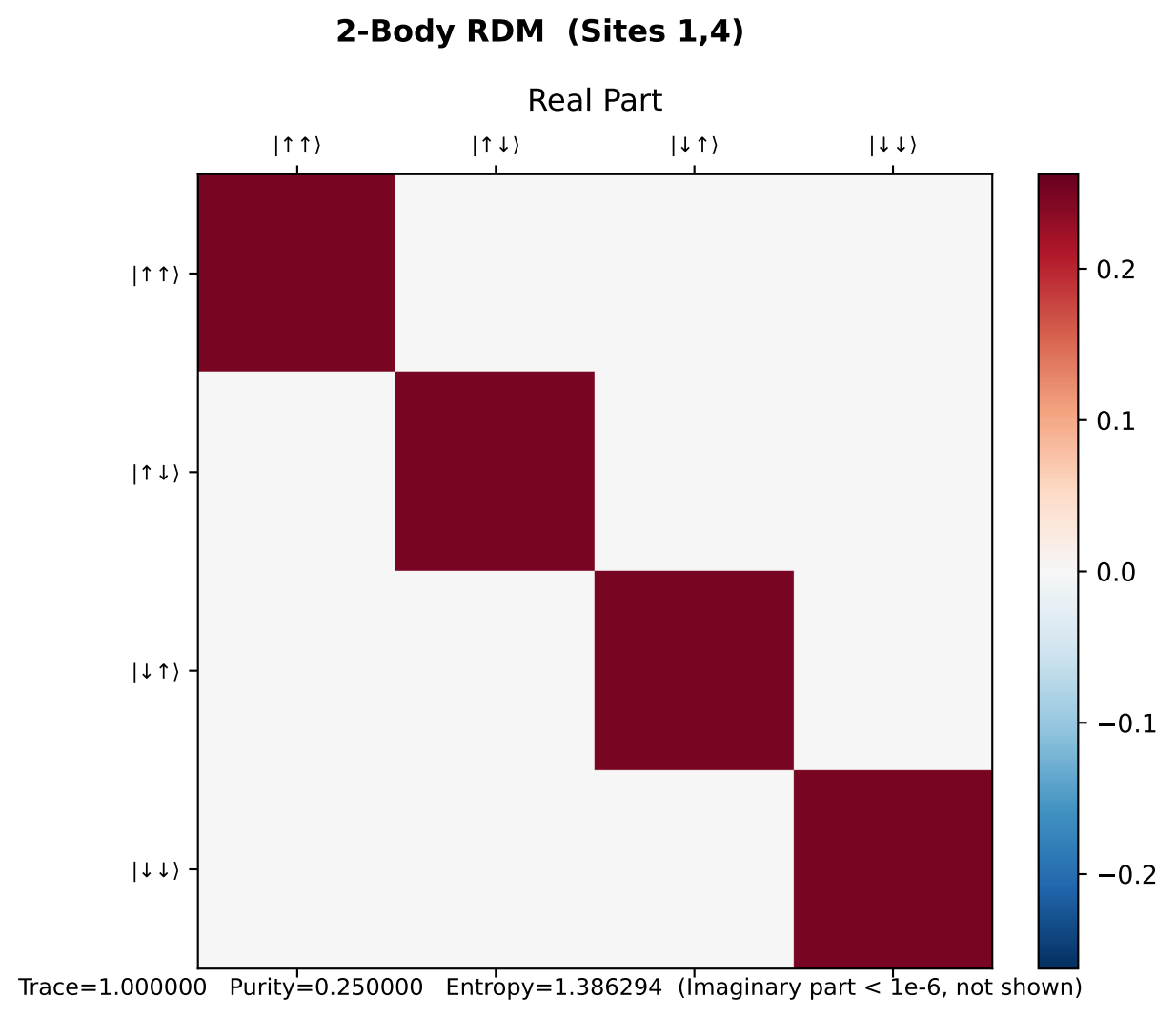
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 1,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.250000 Entropy=1.386294 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 1,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2500000000
Von Neumann entropy	: 1.3862943611
Eigenvalue min	: 2.500000e-01
Eigenvalue max	: 2.500000e-01
Imaginary part	: < 1e-6 (not shown)

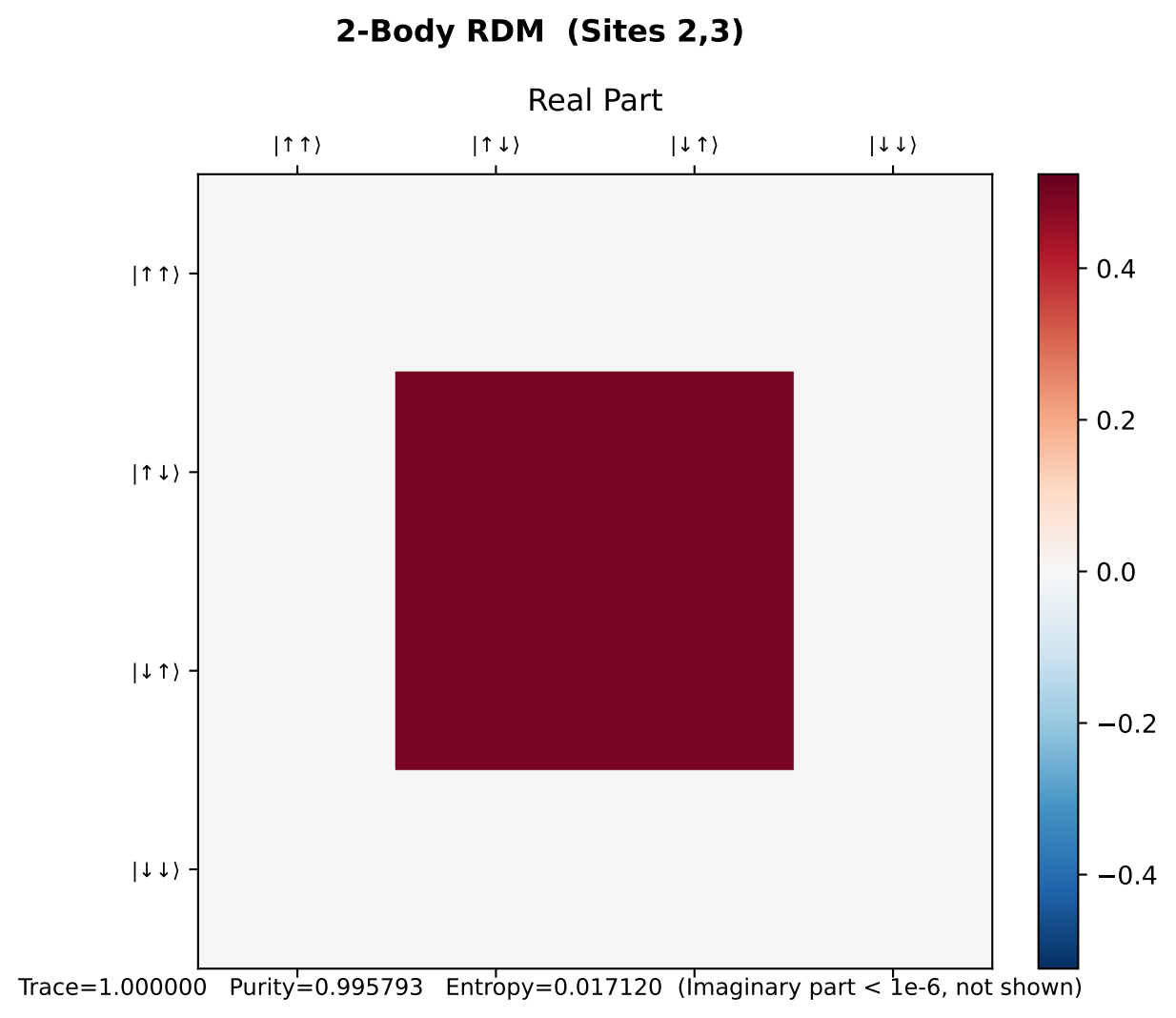
GME Analysis
==========

gamma = Tr(rho^2) = 0.25000000

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 2,3)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.4
0.2
0.0
−0.2
−0.4
Trace=1.000000 Purity=0.995793 Entropy=0.017120 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 2,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.9957931004
Von Neumann entropy	: 0.0171196253
Eigenvalue min	: 2.343160e-04
Eigenvalue max	: 9.978934e-01
Imaginary part	: < 1e-6 (not shown)

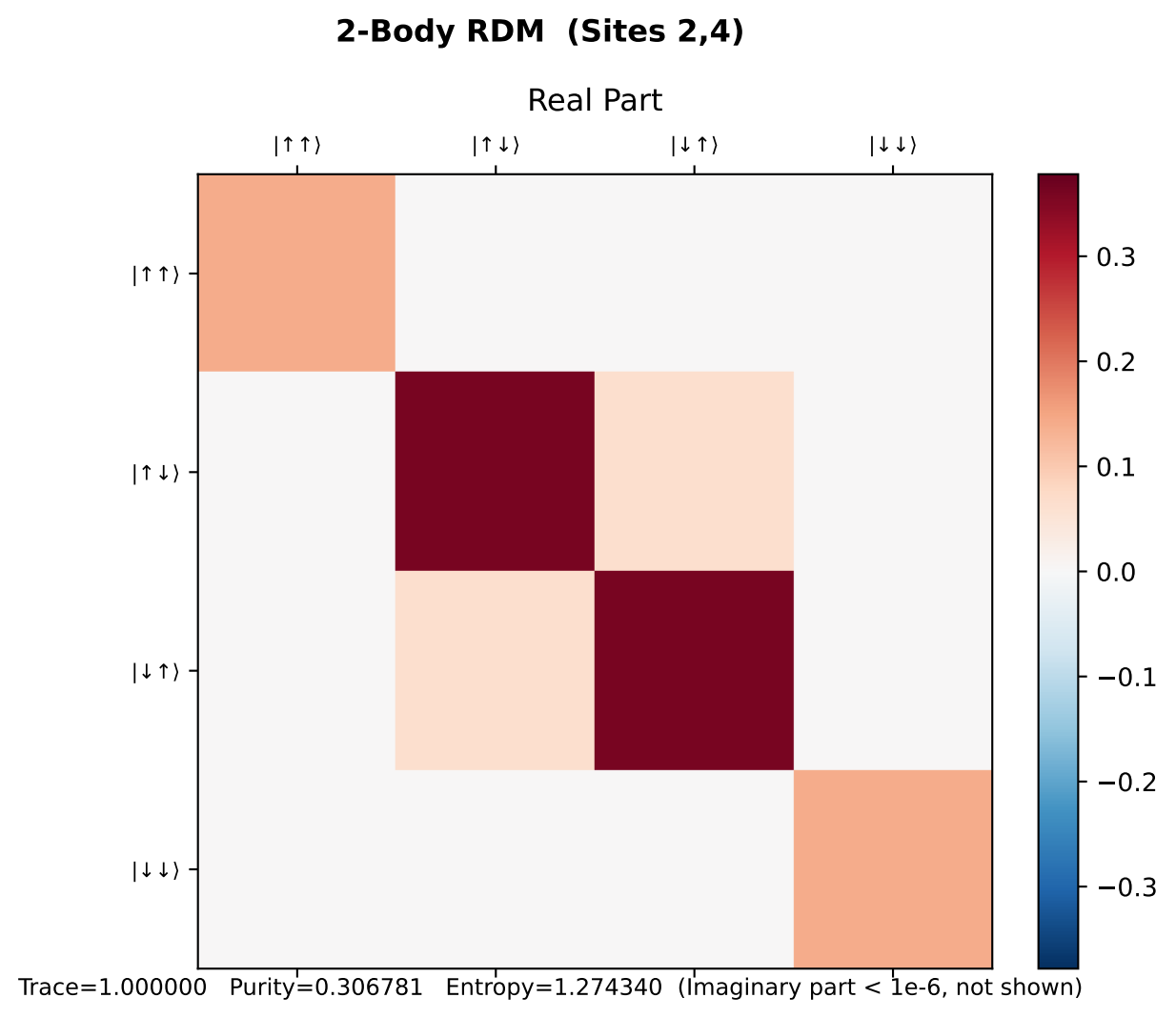
GME Analysis
==========

gamma = Tr(rho^2) = 0.99579310

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 2,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.306781 Entropy=1.274340 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 2,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.3067809701
Von Neumann entropy	: 1.2743395463
Eigenvalue min	: 1.399080e-01
Eigenvalue max	: 4.245124e-01
Imaginary part	: < 1e-6 (not shown)

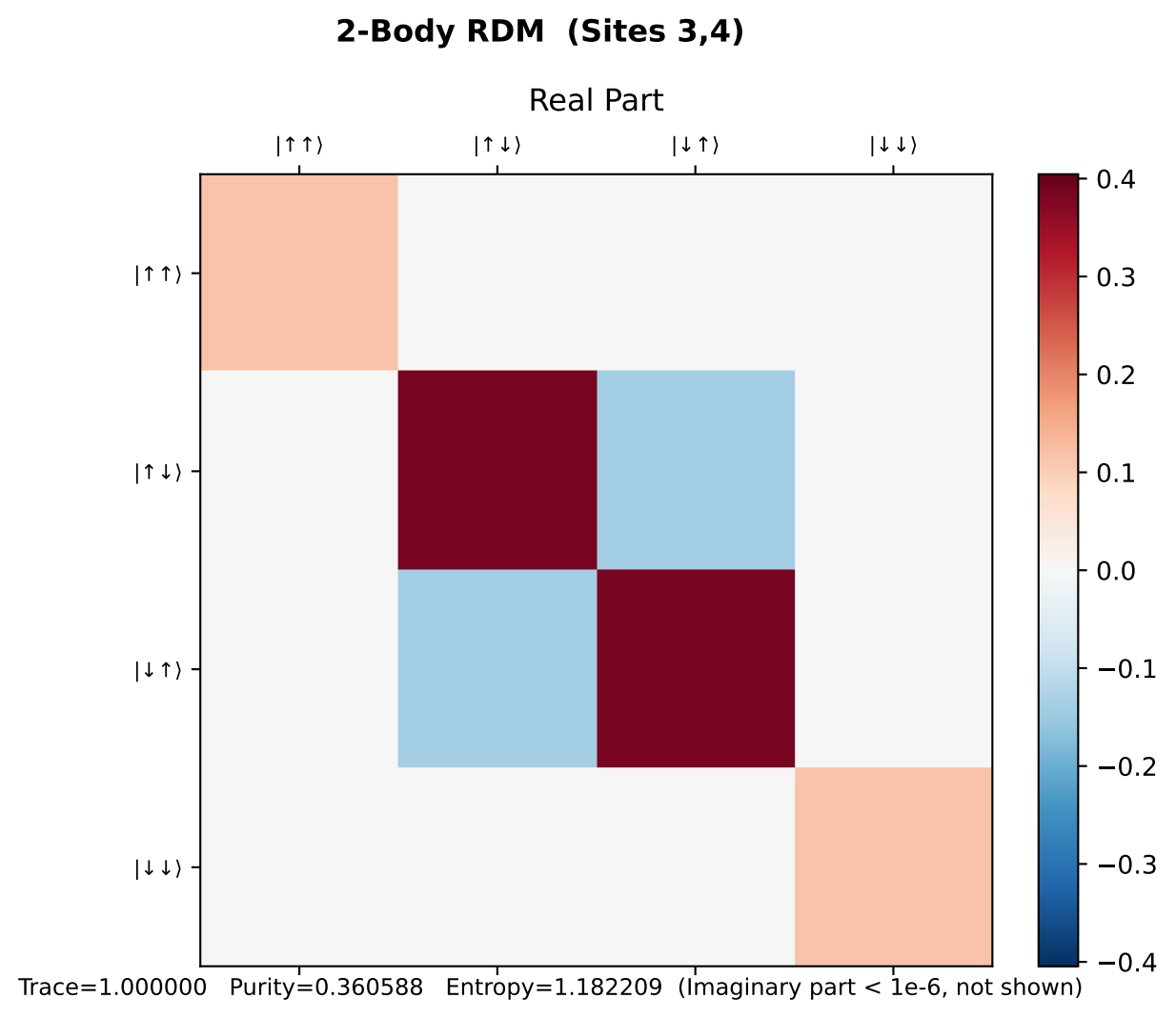
GME Analysis
==========

gamma = Tr(rho^2) = 0.30678097

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


2-Body RDM (Sites 3,4)
Real Part
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
Trace=1.000000 Purity=0.360588 Entropy=1.182209 (Imaginary part < 1e-6, not shown)

Analysis Summary - 2-Body RDM (Sites 3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.3605876311
Von Neumann entropy	: 1.1822085787
Eigenvalue min	: 1.147840e-01
Eigenvalue max	: 5.220629e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.36058763

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.